

Doria Pertea

Baltimore, Maryland • <https://dperte.github.io/swe/>
dgperte@gmail.com • (410) -564-6131

WORK EXPERIENCE

Software Engineer • 05/2022 - Present
Abt Global • Rockville

- Led development of multiple **full-stack enterprise applications** featuring React, TypeScript, Java/Node.js/Python, and Oracle/PostgreSQL, supporting thousands of users across federal clients
- Optimized large-scale data pipelines and **SQL** queries for applications using **Oracle RDBMS**, reducing processing times from **24+ hours to minutes** for multi-million row datasets
- Designed and implemented a **React-based financial modeling tool** integrating a **Python** forecasting engine (via Pyodide and Web Workers), enabling data visualization and analysis for **billion-dollar investment decisions**
- Led the development of a Harvard University **React, TypeScript, Python (FastAPI)**, and **PostgreSQL** application which provides paying U.S. institutions access to meaningful insights from anonymized faculty job satisfaction data
- Modernized a suite of data-intensive applications for the US Army Corps of Engineers, designed to help process, track, and map vessel and cargo movement across US water ways using **React, GraphQL, Java** and **LeafletJS** for **route visualization**
- Rebuilt a legacy **document search platform** in **Angular, NestJS** and **Oracle**, introducing parameterized queries, asynchronous processing, and **scheduled OCR tasks**, to improve document metadata and search relevance by **3x**
- Developed reusable, testable **React** components and contributed to quality assurance, ensuring **Cypress test** coverage for NIH's dietary supplement calculators built with **NextJS** and **TypeScript**
- Engineered a **C# .NET backend** for the EPA, scaling computations from ~3,800 to ~84,000 census regions; introduced multi-threading to reduce execution time by **50%+**
- Acted as primary technical point of contact for stakeholders, translating business requirements into scalable technical solutions
- Maintained and deployed containerized applications to development, staging and production environments using **Docker, AWS**, and Cloud.gov, implementing **CI/CD** pipelines to support reliable and repeatable releases

Full Stack Engineer • 03/2021 - 05/2022
USDA DAWN Project • College Park

- Developed a responsive **React dashboard** using **Material UI, Chart.js**, and **Google Maps API**, allowing farmers to optimize crop output and visualize planting recommendations based on crop type, soil conditions, weather patterns, and field location
- Engineered a data pipeline in **Python Flask** to process large **.netCDF** climate datasets into optimized formats for frontend consumption, improving load times for geographic visualizations
- Built and maintained a **REST API** using **Node.js/Express** and **MongoDB**, later contributing to a migration toward **Go** to improve concurrency and system performance of an agricultural dashboard designed to help farmers optimize their growing seasons

Teaching Assistant • 06/2021 - 08/2021
University of Maryland • College Park

- TA'd for 2 courses at the university: "An introduction to **front end development in React**" and "An introduction to **back end application development in Node and Express**"
- Covered **JavaScript fundamentals** including the event loop, JavaScript's single-threaded concurrency model, synchronous vs. asynchronous behavior, and callback patterns, holding office hours to help debug student projects

Computational Biology Intern • 07/2020 - 08/2020
Johns Hopkins Whiting School of Engineering • Baltimore, MD

- Developed **bash** and **R** scripts to **automate visualization workflows for genomics** researchers, generating **Manhattan plots, imputed state visualizations**, and **CHIP-Seq data** displays for use in scientific research papers and presentations
- Contributed to genomics research to advance personalized medicine by identifying signals in regulatory gene elements with potential to regulate telomere length by creating tools to visualize nucleotide polymorphisms (SNPs) across **large genomic datasets**

EDUCATION

Bachelor of Science in Computer Science
University of Maryland
05/2022

SKILLS

Frontend: React, Javascript, TypeScript, NextJS, HTML5, Ajax, CSS, SCSS, Angular, MaterialUI, JSX, AG Grid, D3.js, Highcharts, Webpack, Yarn, Vite, Frontend Testing Frameworks (Cypress/ Selenium/Jest/React Testing Library), Mapping Libraries (Leaflet/Google Maps Api/Mapbox), Design Systems (Bootstrap/ Telerik/MaterialUI/et al.), Web Accessibility Standards, Google Analytics, AXE Dev Tools

Backend: Node.js, Express, NestJS, Python, Java, PL/SQL, SQLite, C# .NET

Databases: PostgreSQL, Oracle, SQLite, MongoDB, Redis

Cloud & DevOps: Docker, Amazon AWS (S3/EC2), Cloud.gov, Linux, Git, CI/CD (Bitbucket Pipelines/Github Actions), Bash, NPM, NVM, Debugger